

ASSIGNMENT - 1

AMPLITUDE MODULATION WITH AND WITHOUT CARRIER SUPPRESSION (ED-2960) TELECOMMUNICATION TRAINER

Object

1. To demonstrate amplitude-modulated signals with and without the suppression of the carrier.
2. To demonstrate the saving of power without loss of information, which results from suppressing the carrier.

Equipment required

- Double-beam oscilloscope, 5MHz bandwidth.
- Low-frequency generator (0.1Hz).
- Double side-band Module 2960C.
- Power supply : ED-2900P

INTRODUCTION

You know how conventional amplitude modulation impresses a signal, representing information to be transmitted, on a carrier frequency signal, which is simply a means for facilitating transmission. You know also how the result is a composite signal made up of the carrier, and upper and lower sidebands, of higher and lower frequency than the carrier.

For a signal $s=S\sin\omega_s t$ impressed on a carrier $c=C\sin\omega_c t$ the composite result is

$$y = C \sin\omega_c t + \frac{1}{2} SC [\cos(\omega_c - \omega_s)t - \cos(\omega_c + \omega_s)t]$$

In the expression on the right-hand side, the first term is simply the carrier. Notice that if S were zero, so that there was no signal, that term would still be there. It does nothing therefore to tell us whether there is any information present or not: despite its name it does not (by itself at least) carry any information.

The second and third terms do depend on S and on s , the two characteristics of our simple signal. In the case of a complex signal, more sidebands would be present, all dependent on the nature of the signal, and therefore contributing to the information.

In this assignment you will see that it is possible to generate the sidebands in a transmitter without having to generate the carrier (except as a low-level auxiliary signal at the input to the modulator). You will see also that there is a considerable power saving in so doing.

In order to demonstrate this, a modulator is needed which can perform the modulation equally well with or without the carrier. The 2960C module uses an integrated circuit modulator in a mode which causes the output to be simply the product (result of multiplying together) of its two input signals. One input being the carrier $c = C \sin \omega_c t$, and the other a signal s , the output is $sC \sin \omega_c t$.

If the signal $s = S \sin \omega_s t$ is applied to this modulator, the resultant output signal is

$$sc = SC \sin \omega_s t \sin \omega_c t = \frac{1}{2} SC [\cos(\omega_c - \omega_s)t - \cos(\omega_c + \omega_s)t]$$

This represents just the two sidebands of the composite modulated signal y with which we started the assignment.

If we want to obtain the complete signal y , then the carrier term $C \sin \omega_c t$ must be added. This is very easily done, for all that is needed is to add 1 (a steady d.c) to the signal fed into the modulator. In the 2960C the 'set carrier' control adjusts the level of the output carrier by feeding an adjustable D.C. signal into the modulator, along with the signal from the signal generator.

The tuned power amplifier following the modulator is in effect a small radio transmitter. It could be connected to an aerial to radiate a signal (although this would be illegal). However, in order to show you the power which it generates, the transmitter is instead connected to a small lamp, which lights with a brightness which depends on the power fed to it. This lamp is the 'RF power monitor' on the module.

PRACTICAL 1.1

Modulation with and without carrier suppression

Plug the Module 2960C into the power supply, then switch on the supply.

Adjust the 'set carrier control' to extinguish the 'RF power monitor' lamp. This reduces the transmitter's carrier output to zero (nominally).

Connect the Y1 channel of the oscilloscope to the output of the 1MHz oscillator in the module. Set the sensitivity of this channel to 1V per division, allowing for any attenuation introduced by the probe. Set the time base to give a scale of 1s per division, and connect this channel to the 'RF monitor point' on the module.

Slowly rotate the 'set carrier' control counterclockwise until a 1MHz signal of approximately 10V peak-to-peak amplitude is present at the RF monitor point (Y2 channel). Observe, and make a careful note of, the phase relationship between the two displayed waveforms.

Next, slowly rotate the 'set carrier' control clockwise, so that the signal at the monitor point decreases, passes through a null and then increases once more. Set the voltage to about 10V peak-to-peak. Observe the phase relationship between the two waveforms in this new state. and make a note of your observation.

Q 1.1 Taking the 1MHz oscillator signal (on Y1) as a reference, how much did the phase of the RF monitor signal change when you moved the 'set carrier' control?

Q 1.2 Why do you think this phase change took place?

Readjust the 'set carrier' control for a null output once more. use the oscilloscope to detect the null precisely. Connect the low-frequency generator to the 'modulation input' terminals. Set its controls to give an output of 5V peak-to peak at 0.4Hz. Rotate the 'set carrier' control clockwise until the signal on the 'RF monitor' point reaches a maximum peak-to-peak value of about 15V. Observe the phase of the RF output changing regularly in synchronism with the modulating input from the oscillator. The oscillator is now having the same effect as your action on the 'set carrier' control formerly had. Raise the frequency of the oscillator to 4Hz, and you will be able to see the process speeded up.

To show better how the wave is modulated, choose a very slow time-base speed, giving a scale of say 20 ms/cm. This is the same signal as you looked at before, but on a vastly different time scale. Transfer the Y1 lead of the oscilloscope to the modulating oscillator. You should now have a display resembling Fig. 1-1. It may be necessary to readjust the trigger controls of the time-base to restore the display.

The flicker of the oscilloscope is tiring to watch, so increase the modulating frequency to 400Hz, and reset the time-base to 200 μ s/cm. The display will again resemble Fig. 1-1 but should be quite steady: if not, try adjusting the trigger controls of the oscilloscope.

What you now have is a "double-sideband suppressed-carrier' signal, or DSSC signal for short. It is the result of simply multiplying the carrier oscillation and the modulating signal together. The product does not contain any carrier frequency component.

If the modulation disappears, so does the output signal. Turn the 'set modulation' control fully anti-clockwise, and the output signal will disappear.

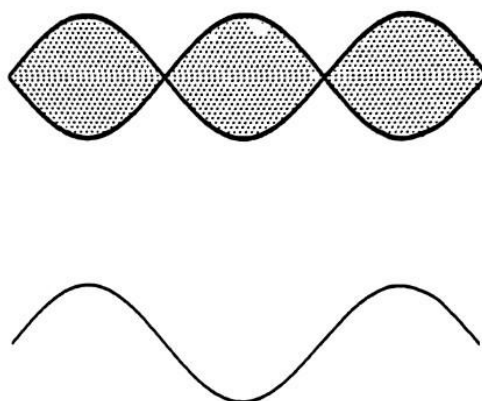


Fig. 1-1 Modulating and DSSC modulated signals

Make a copy of Fig. 1-1, and on it mark the waveform you would expect to obtain if the DSSC signal were applied to an ordinary AM detector, such as Fig. 1-2.

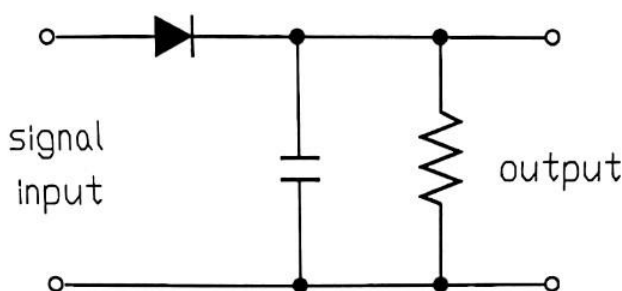


Fig. 1-2

Q 1.3 Would you expect this detector to work satisfactorily on DSSC signals? Explain your answer.

With the 'set modulation' control still fully anti-clockwise, adjust the 'set carrier' control until the RF output is about 8V peak-to-peak.

This RF output represents an unmodulated carrier.

This carrier may now have conventional amplitude modulation applied to it, by turning the 'set modulation' control clockwise. 100% modulation, as shown in Fig. 1.3, is reached when the maximum RF is about 16V peak-to-peak, and the signal goes down to zero amplitude at one point in the cycle. (Note that normal AM modulators limit at zero amplitude, and do not allow the phase to reverse, as you have seen happens with this more versatile one). Leave the equipment set for 100% modulation, in readiness for the next Practical.

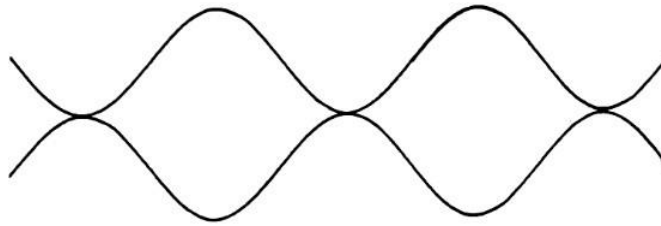


Fig. 1-3

Note that the problems which occur when a DSSC signal is applied to a simple detector do not apply in this case (i.e. when the carrier is suppressed). so that suppressed-carrier signals require more complicated detectors.

It turns out that a detecting system for DSSC signals is very much more complicated indeed. Why then is suppression of the carrier so desirable?

The answer is largely a matter of power.

PRACTICAL 1.2

Power required for AM and DSSC transmissions

Your apparatus should still be set up with an RF signal 100% modulated at 400Hz. The power in the signal is being fed into the lamp on the panel of the module, marked 'RF power indicator', which should be glowing. Watch the lamp while you turn off the modulation, by turning the 'set modulation' control fully anti-clockwise. Notice that the power is reduced very little. Most of the power in the modulated wave is in the constant carrier (which conveys no information, since no change in the modulating signal alters it).

Set the modulation back to 100%, and look at the outline, or envelope, of the RF output on the oscilloscope. The detector follows just the upper outline, so that in this case the audio output would be 8V peak-to-peak.

Remove the modulation, by turning the 'set modulation' control. Then set the carrier output to zero. Here you have a DSSC system with no modulating signal to be sent. The system puts out no signal, as you can see from the Oscilloscope and from the lamp, and the power output is zero. The power required to drive the system is thus also (in principle) zero when no signal is present.

Now apply a modulating signal, and adjust it to the level which gives an output of 16V peak-to-peak once more. Notice that the bulb is slightly brighter than it was before with the 100% modulated wave with carrier present. This means that extra power is obtainable from the same equipment, since clearly the peak values of voltage and current are the same in both cases. What you have set up now represents the maximum signal which can be sent on suppressed-carrier basis from the same basic transmitter.

What good does the extra power do? If we consider the envelope to be that outline of the signal which would be followed by a detector reproducing the original signal, you can see that it is now 16V peak-to-peak, instead of the 8v which applied to the previous 100% modulated wave. Thus the voltage which would be generated by a detector capable of distinguishing positive and negative phases would be doubled: the received power would be quadrupled.

Summing up, we have so far discovered that we can radiate more power, and because we waste none, an enormous increase in the power associated with the signal (or information to be sent) is achieved.

This is not the whole advantage. Signals such as speech and music vary widely in power all the time. The level at which such signals are transmitted has to be such that, even if the largest peaks of the signal are clipped off, the periods when the signal has its larger values do not give rise to severe distortion; these values must be within the maximum output capability of the transmitter. It follows that much of the time the transmitter signal is appreciably less than maximum.

Look once again at the power indicator bulb, with the signal still at 16V peak-to-peak. Then reduce the amplitude to 8V peak-to-peak. Notice how much the power output is reduced. It is less than the power in the unmodulated carrier, i.e in the conventional AM signal with no signal.

In the practical case therefore, transmission with suppressed carrier not only enables a stronger information signal to be sent for a given maximum power, but it significantly reduces the average power.

PROCEDURE

The setup of the experiment is shown in figure (1)

- a. Connect DS (2960C) to ED-2960 console.
- b. Use function Generator to generate 1Volt (p-p) sinusoidal signal with frequency of 1KHz and check it using the oscilloscope.
- c. Connect 1 MHz oscillator on DS 2960C) to oscilloscope to measure amplitude and frequency of the carrier.
- d. Connect the output of the function generator to the 'modulation input' of the DS (2960C) and slowly rotate the 'set modulation' to the right.
- e. Connect the output of the RF monitor to oscilloscope in order to display the modulated signal.
- f. Slowly rotate the set carrier to generate and plot the following:
 1. Double side-band with transmitted carrier ($m > 1$)
 2. Double side-band with suppressed carrier ($m = 1$)

3. Double side-band with suppressed carrier ($m < 1$)
- g. Using the formula for m given in the discussion part calculate the modulation index for the three modulated signals given in part f
- h. Using the function generator change the signal shape to triangular and note the modulated signal on the oscilloscope.

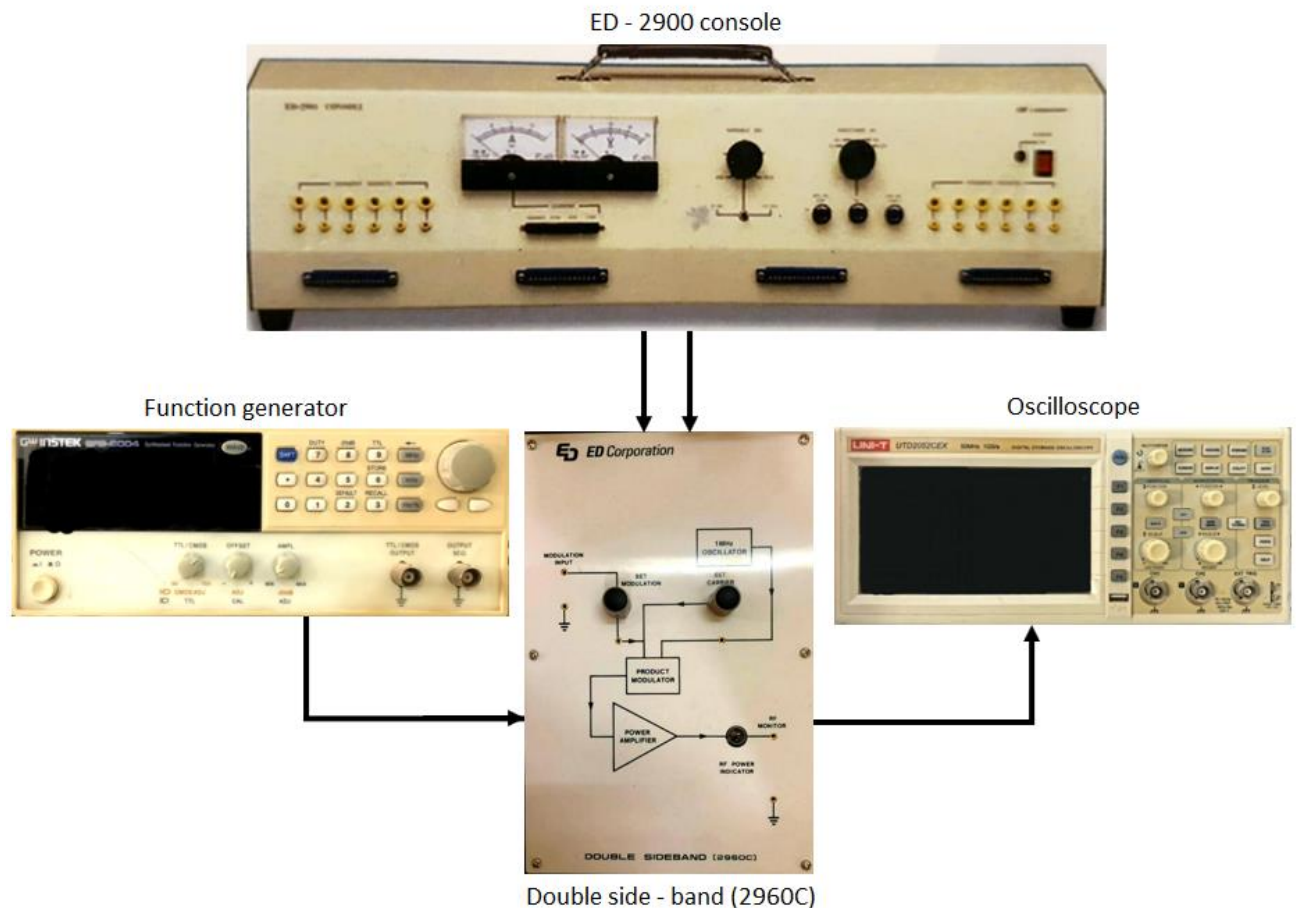


Figure (1)

DISCUSSION

- 1- Set the carrier and observe the output at RF monitor. Discuss three cases about modulation index.
- 2- Give the significance of modulation index?
- 3- Change the input to square wave and draw the modulated waveform.
- 4- Compare between AM wave, DSB-SC, and SSB-SC.
- 5- Explain how AM wave is detected?

ASSIGNMENT 2

AMPLITUDE MODULATION AND DEMODULATION; SINGLE-SIDEBAND

Object

1. To demonstrate the process of sending and receiving a single-sideband signal.
2. To demonstrate the importance of synchronizing the carriers at the sending and receiving ends.

Equipment required

- Dual-beam oscilloscope
- Audio-frequency signal generator
- Digital frequency counter, or equivalent
- Single-sideband module 2960D
- Power supply: ED-2900P

INTRODUCTION

Assignment - 1 showed how, for a given peak transmitter power, suppressed-carrier transmission offered more information-carrying power at the output, for less consumption of power. The problem was the difficulty in detecting signals consisting simply of the two sidebands. This assignment is concerned with the most usual method of securing the advantages of suppressed carrier working, without the worst of the problems. That method is to transmit only one of the sidebands: the carrier and the other sideband are both suppressed. The single-sideband technique starts with the generation of a suppressed-carrier signal such as you met in Assignment - 1. And then removes one of the sidebands from it. Not only does this remove problems which you have met, but it also introduces a further economy – in bandwidth. This is a most important economy in all forms of communication. If radio transmitters occupy half the bandwidth, twice as many of them can work in a given frequency band without mutual interference. If with two sidebands a telephone channel is say 8kHz wide, then a 40kHz-wide channel can carry five channels; if the channel width is halved by shedding one sideband. it can carry ten. The result is a huge saving in cost.

The 2960D Single-sideband Module contains in effect a single-sideband sending station connected directly to a corresponding receiving station.

There are also auxiliary amplifiers for a microphone and speaker. There are different methods of generating single-sideband signals, but the 2960D Module uses the most straight forward.

If you look at the panel of the module, you will find at the top left a pair of audio input terminals, into which the original signal is connected.

Considering it to be a single-frequency signal, its spectrum is shown at Fig. 2-1(a). The signal is fed to a product modulator, whose other input comes from a 100kHz oscillator. The operation of the modulator is exactly like the production double-sideband suppressed-carrier (DSSC) signals in Assignment 1. The modulator is permanently set to suppress the carrier. The output of the modulator therefore consists just of two sidebands, (Fig. 2-1(b)) of frequency $(100 \pm f)$ kHz, where f kHz is the audio frequency.

This signal is passed into a filter which removes the unwanted sideband (Fig. 2-1(c)). The output of this filter is the signal in the form in which, in a communication system, it would be sent to the line, or the power amplifier of a transmitter.

The 'mixer' which the signal enters at the receiver is another multiplier, or modulator. You may notice its similarity to the frequency changer in the superhet receiver: it is indeed a frequency changer too, although in this application it changes the incoming frequency not to an 'intermediate frequency', but all the way back to the original audio frequency.

This product modulator, like the others you have met, on receiving a signal $s = S \sin 2\pi f_s t$, and from the local oscillator a carrier $c = C \sin 2\pi f_c t$, produces an output

$$sc = \frac{1}{2} SC [\cos 2\pi(f_c - f_s)t - \cos 2\pi(f_c + f_s)t]$$

In this case the signal is of frequency

$$(100 - f) \text{ kHz, so } f_s = 100 - f$$

In a properly adjusted system, the carrier frequency f_c will be equal to the carrier frequency originally used at the sending end. In this case it should therefore be 100 kHz.

Substituting $(100 - f)$ and 100 respectively for f_s and f_c in the expression for the receiver's modulator output, the output becomes

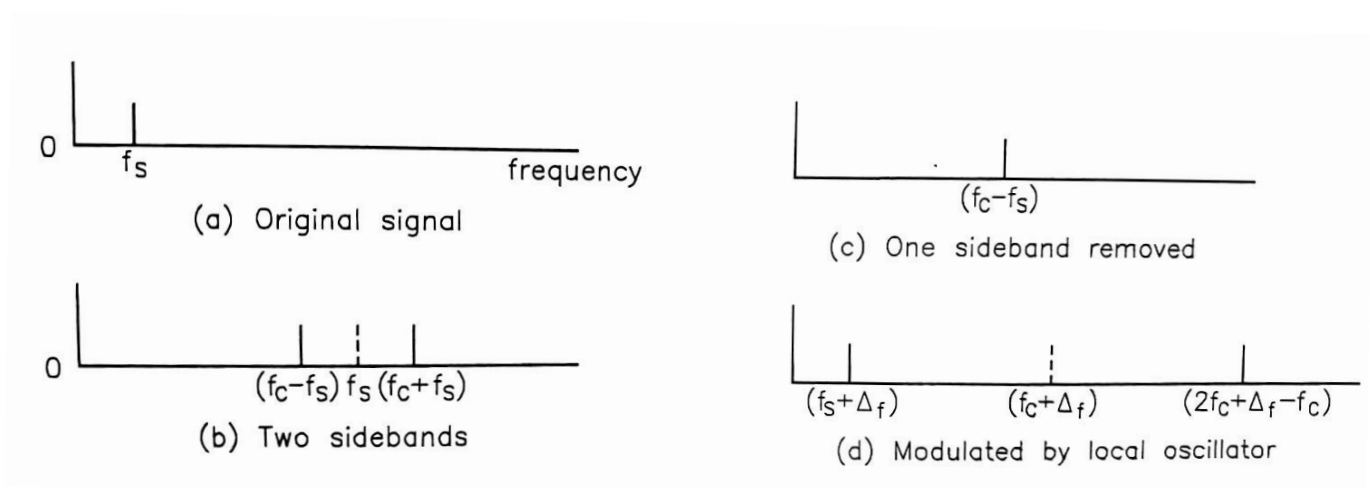
$$\begin{aligned} sc &= \frac{1}{2} SC [\cos 2\pi(100 - 100 + f)t - \cos 2\pi(100 + 100 - f)t] \\ &= \frac{1}{2} SC [\cos 2\pi f t - \cos 2\pi(200 - f)t] \end{aligned}$$

The first of these terms is the original frequency (audio frequency in this case), which passes through the low-pass filter to the audio output terminals. The second term is of high frequency, and is rejected by the filter.

Fig. 2-1(d) shows the two terms slightly shifted by an assumed error Δf in the local oscillator frequency. You may notice that although the original signal frequency is regained, the cosine, replacing the sine we started with, implies a change of phase. We could just as easily have started with the local oscillator represented by $c = C \cos 2\pi f_c t$, which would have given us the original phase.

Quite often however the phase is not important, as you will see. The situation is however quite different from the double-sideband case, where the outputs from the two sidebands, each rotated 90° in the opposite direction from the other, could cancel one another out.

Fig. 2-1 Frequencies present at various stage of single sideband transmission



PROCEDURE

The setup of the experiment is shown in figure 2-2

- Connect single side-band DS(2960D) to ED-2900 console.
- Use the function generator to generate 1 Volt(p-p) sinusoidal signal with frequency of 1KHz and check it using the oscilloscope.
- Connect 100 KHz oscillator on DS(2960D) to oscilloscope to measure amplitude and frequency of the carrier.
- Use two channels of the oscilloscope, one of them by connecting audio input pin DS(2960D) to oscilloscope and the other one for transmitter medium pin DS(2960D), then display both signals.
- On board DS(2960D), connect local oscillator pin to oscilloscope and adjust the frequency to 100 KHz to match the carrier frequency.
- On board DS(2960D), connect the input and output of the low pass filter to the two channels of the oscilloscope then compare the results signals.

- g. On board DS(2960D), connect the audio input pin and audio output pin to the two channels of the oscilloscope then compare both signals.

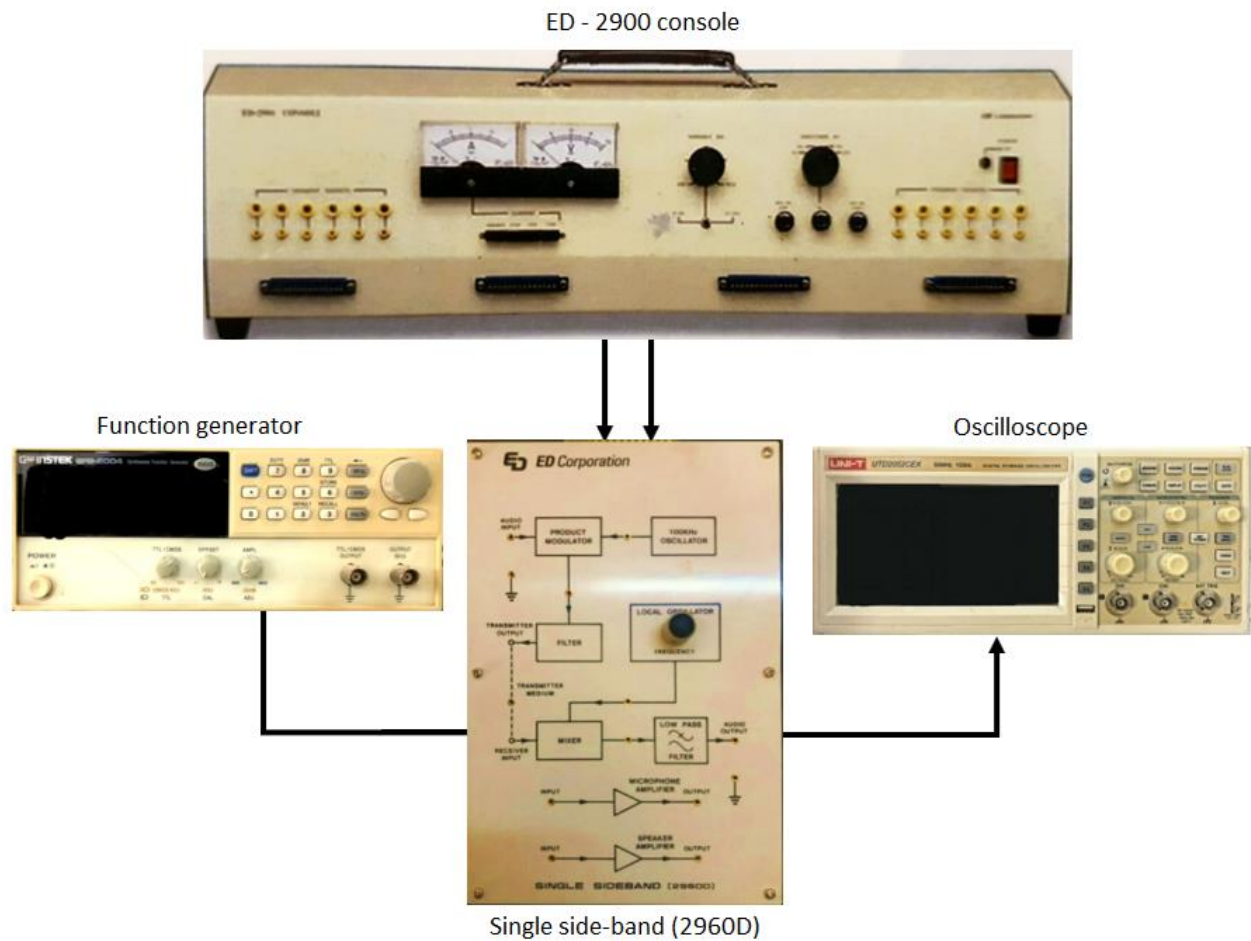


Fig. 2-2

DISCUSSION

- 1- What are the different methods to generate SSB-SC signal?
- 2- What is the advantage of SSB-SC over DSB-SC?
- 3- Why SSB is not used for broadcasting?
- 4- Explain Phase Shift method for SSB generation.

ASSIGNMENT - 3

3.1 GENERATION OF PULSE AMPLITUDE MODULATION (PAM) SIGNAL

Object

1. To understand the generation of PAM signal after natural and flat-top sampling.
2. To learn the influence of frequency of sampling signal in time-domain on PAM signal.

Equipment required

- Power Supply(U-2920A)
- Signal Source(U-2920B)
- Pulse Amplitude Modulation(U-2920C)
- Digital Storage Oscilloscope (2-CH, 60[MHz])

PROCEDURE

1. Prepare module and measuring device, U-2920A, U-2920B, U-2920C and Oscilloscope. Set all ADJ. control terminal to MIN and provide power to devices by connecting U-2920B and U-2920C to power supply (U-2920A).
2. Connect TRIANGLE output of SIGNAL SOURCE to J2 terminal of PAM module, then connect CLK terminal of SIGNAL SOURCE to CLK PULSE terminal of PAM module and set FREQUENCY SELECTOR of SIGNAL SOURCE to 32[kHz], 16 [kHz] and 8 [kHz]
3. Set oscilloscope to as following: TIME/DIV to 50[μs], DC INPUT COUPLING, CH-1 VOLT/DIV to 2[v] and CH-2 VOLT/DIV to 1[v]
4. Connect CH-1 input probe of oscilloscope to J2 terminal of PAM module, set frequency to 3.2[kHz] by adjusting Frequency ADJ of AUDIO GENERATOR in SIGNAL SOURCE, and set amplitude to 4[V_{p,p}] by adjusting Frequency Amplitude ADJ.
5. Set SAMPLING MODE Switch SW1 to NATURAL MODE (natural sampling) in PAM module and connect J3 terminal and J4 terminal. Then connect CH-2 input probe of the oscilloscope to J6 terminal of PAM module and plot the waveform for each FREQUENCY SELECTOR of SIGNAL SOURCE mentioned in 2.
6. Set SAMPLING MODE Switch SW1 to FLAT-TOP mode (flat-top sampling) in PAM module and repeat 5 then plot.

7. Connect CH-1 input probe of oscilloscope to TP1 of PAM module and connect CH-2 input probe of oscilloscope to TP2 of PAM module, plot the waveform of message signal and sample-retention signal if FREQUENCY SELECTOR of SIGNAL SOURCE is set to 32[kHz], 16[kHz] and 8[kHz].

3.2 PAM SIGNAL DEMODULATION

Object

1. To define sampling theory and Nyquist rate.
2. To learn the fact that message signal can be recovered from PAM signal by using Low-pass filter.
3. To understand principle to generate aperture distortion from flat-top sampling.

Equipment required

- Power Supply(U-2920A)
- Signal Source(U-2920B)
- Pulse Amplitude Modulation(U-2920C)
- Digital Storage Oscilloscope (2-CH, 60[MHz])

PROCEDURE

1. Prepare module and measuring device, U-2920A, U-2920B, U-2920C and Oscilloscope. Set all ADJ. control terminal to MIN and provide power to devices by connecting U-2920B and U-2920C to power supply (U-2920A)
2. Connect SINE Output terminal and CLK Output terminal of SIGNAL SOURCE to AUDIO INPUT and CLK PULSE terminal of PAM module. Then, connect J1 with J2 terminal and J3 with J13 of PAM module.
3. Set oscilloscope to as following: TIME/DIV to 0.2 [ms], CH1 and CH-2 VOLT/DIV to 2[v].
4. Connect CH1 input probe of oscilloscope to J1 terminal of PAM module, set frequency to 2[kHz] by adjusting Frequency ADJ. of AUDIO GENERATOR in SIGNAL SOURCE, set amplitude to 4[Vp.p] by adjusting Amplitude ADJ.
5. Set FREQUENCY SELECTOR in SIGNAL SOURCE to 8[kHz] and set SAMPLING MODE Switch SW1 in PAM module to NATURAL mode (natural sampling).
6. Set ORDER SELECTOR Switch SW2 to 2nd and connect CH2 input probe of oscilloscope to AUDIO OUTPUT 1 of PAM module, then plot the resulting waveform of PAM demodulation.

7. Set ORDER SELECTOR switch SW2 to 4th and connect CH2 input probe of oscilloscope to AUDIO OUTPUT 2 of PAM module, then plot the resulting waveform of PAM demodulation.

Discussion

1. What is the minimum rate at which a speech signal can be sampled for the purpose of PAM?
2. Which is better, natural sampling or flat topped sampling and why?
3. Is PAM & Demodulation is sensitive to Noise?
4. Give an application for PAM.

ASSIGNMENT - 4

GENERATION OF PTM SIGNAL

Object

1. To understand method to generate PTM (Pulse Time Modulation) signal.
2. To understand the characteristics of PTM signal in time domain.
3. To understand the relationship between PWM and PPM signal.

Equipment required

- Power Supply(U-2920A)
- Signal Source(U-2920B)
- Pulse Time Modulation (U-2920D)
- Digital storage Oscilloscope (2-CH, 60[MHz])

PROCEDURE

1. Prepare module and measuring device and set all ADJ. control terminal to MIN respectively and provide power to devices.
2. Set oscilloscope to as following:

TIME/DIV	0.2[ms]
CH-1 & CH-2 VOLT/DIV	2 [V]
TRIGGER MODE	AUTO
TRIGGER SOURCE	CH-1
VERTICAL MODE	DUAL
INPUT COUPLING	DC

3. Connect SINE, CLK and RAMP output terminal of SIGNAL SOURCE to AUDIO INPUT, CLK PULSE and RAMP INPUT terminal of PTM module respectively.
4. Set FREQUENCY SELECTOR of SIGNAL SOURCE to 8[kHz]
5. Connect CH-1 input probe of oscilloscope to TP1 of PTM module, set frequency to 1[kHz] by adjusting Frequency ADJ. of AUDIO GENERATOR and amplitude to 4[V_{P-P}] by adjusting Amplitude ADJ.
6. Connect CH-2 input probe of oscilloscope to TP3 of PTM module and adjust Amplitude ADJ. of RAMP GENERATOR to ramp voltage be 5[V_{P-P}].
7. Remove CH-1 input probe of oscilloscope connected with TP1 of PTM module and then connect it to TP4. And then plot the waveform that would be appeared on oscilloscope screen.

6. Use Oscilloscope to plot the following waveforms:

No.	CH-1	CH-2
Fig.1	Sin-1V/DIV- 200 μ sec	Sample&hold -1V/DIV - 200 μ sec
Fig.2	Ramp – 2V/DIV – 100 μ sec	
Fig.3	Sample&hold -1V/DIV - 200 μ sec	Ramp – 1V/DIV – 100 μ sec
Fig.4	OUT FROM -2V/VID- 200 μ sec	PWM – 5V/DIV - 200 μ sec
Fig.5	Sin-1V/DIV- 100 μ sec	PWM – 5V/DIV - 100 μ sec
Fig.6	Sin-1V/DIV- 100 μ sec	PPM – 5V/DIV - 100 μ sec

.Discussion

1. What is the advantages of PWM?
2. Give an application for PWM?
3. Compare between PWM and PPM modulation.
4. What is the disadvantage of PWM?

ASSIGNMENT - 5

AID&DIA CONVERSION

Object

1. To understand converting process of Analog-Digital and Digital-Analog
2. To understand the method of counter type AD conversion.
3. To understand the method of parallel D/A conversion.

Equipment required

- Power Supply(U-2920A)
- Signal Source(U-2920B)
- Pulse Code Modulation I (Counter-Type U-2920E)
- Digital Storage Oscilloscope (2-CH, 60[MHz])
- Digital Multimeter (EDM4760)

PROCEDURE

1. Prepare module and measuring equipment and set all ADJ control terminals to MIN respectively and provide the power to all devices.
2. Set frequency selector in Signal source to 8[kHz] and connect CLK, RAMP, SINE terminal output of signal source to CLK, RAMP, AUDIO INPUT terminal of PCM I module.
3. Connect 32[kHz] of signal generator from signal source to CLKx4 terminal and connect 128[kHz] to CLKx16 terminal of PCM I module.
4. Set oscilloscope to as following:

TIME/DIV	0.2[ms]
CH-1 & CH-2 VOLT/DIV	5 [V]
TRIGGER MODE	AUTO
TRIGGER SOURCE	CH-1
VERTICAL MODE	DUAL
INPUT COUPLING	DC

5. Connect CH-1 input probe of oscilloscope to TP1 of PCM I, set the frequency to[1kHz] by adjusting frequency ADJ. control terminal of audio generator and set amplitude to [8Vp-p] by adjusting Amplitude ADJ. control terminal.
6. Connect CH-1 input probe of oscilloscope to TP2 of PCM I, set ramp voltage to [9 Vp-p] by adjusting Amplitude ADJ. control terminal in ramp generator.

7. If CH-1 and CH-2 input probe of oscilloscope would be connected to TP1 and J1 of PCM module, plot the two waveforms would be appeared.
8. Connect J1 terminal and J2 terminal of PCM I module. Then remove CH-2 input probe of oscilloscope from J1 and it would be connected to TP3 of PCM I module. At that time, set TIME/DIV of oscilloscope to 0.1[ms] then plot the two waveforms would be appeared.
9. Remove AUDIO INPUT of PCMI module connected to SINE input of SIGNAL SOURCE temporarily and also remove connection line connecting J1 terminal and J2 terminal temporarily. Then connect DC VOLTAGE output terminal of POWER SUPPLY to J2 terminal of PCM I module.
10. Make lighting status of ENCODER DISPLAY Diode to equal with table below by adjusting DC VOLTAGE ADJ. control terminal. At that time, measure the voltage range of J2 terminal and record.

DC Input Value	Bright of LED
-5	
0	
+5	

11. From now on, we will perform the test on D/A conversion and PCM demodulation. Connect AUDIO INPUT of PCM I module to SINE output of SIGNAL SOURCE again, remove DC VOLTAGE terminal connected with J2 terminal, and then connect J1 terminal and J2 terminal again.
12. Connect SYNC. CLKx4, and PCM OUTPUT of DECODER to SYNC. CLKx4, and PCM INPUT respectively. And then connect J3 terminal and J4 terminal
13. If we set INPUT COUPLING of oscilloscope CH-1, 2 to AC and connect each probe to TP1 and AUDIO OUTPUT of PCM I module, plot waveforms would be appeared in oscilloscope screen. At that time, set TIME/DIV of oscilloscope to [50 μ s]. Compare and observe input message signal and demodulation message signal.

Discussion:

- 1- What is the expression for transmission bandwidth in a PCM?
- 2- What is the expression for quantization noise in a PCM?
- 3- Mention some applications of PCM system.
- 4- What are the advantages of the PCM?
- 5- What are the disadvantages of the PCM?

ASSIGNMENT - 6

PCM- TDM MULTIPLEXING /DEMULTIPLEXING

Object

1. To understand the converting process of analog signal being sampled, quantized according to time slot is converted into digital signal through the experiment.
2. To explain Demultiplexing where PAM signals output from Time Slot Counter and DA Converter are realigned by Sequential Switch.

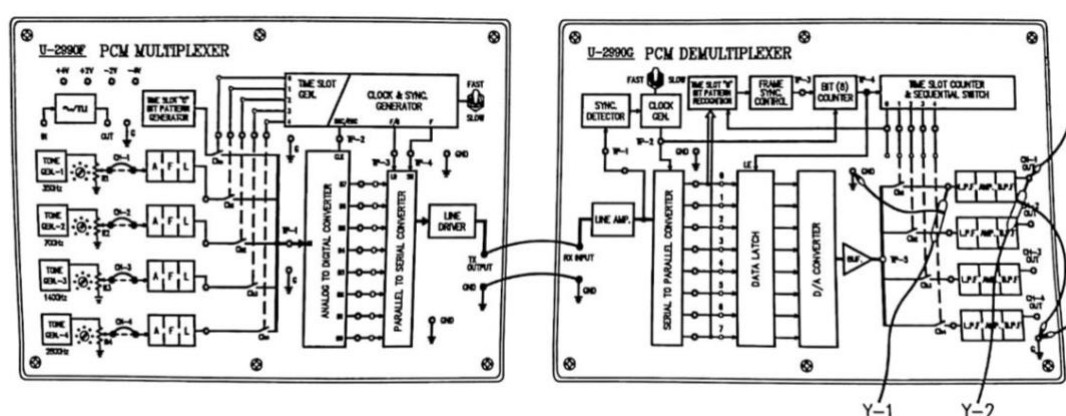
Equipment required

- AF Generator (10Hz - 100kHz)
- Frequency Counter (10Hz - 10MHz)
- Oscilloscope (20MHz, Dual Trace)
- Experimental Module (U-2990F, U-2990G & U-2990E)
- Power Supply Module (U-2990P)

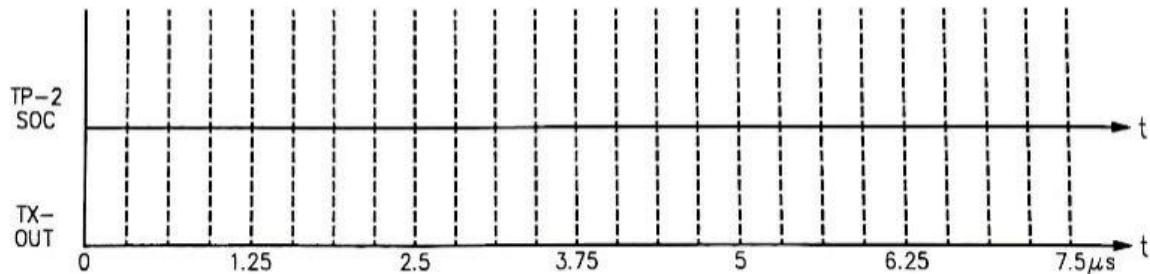
PROCEDURE

The setup of the experiment is shown in figure below

1. With the power switch off, Connect +4V, +2V, -2V, -4V to inputs of each channel and set the clock switch mounted on right top of experimental unit to fast.
2. At first, connect Y-1 probe of oscilloscope to TP-4, connect Y-2 Probe to TP-1 respectively and turn the power switch on. After getting the vertical synchronizing captured by Y-1 input, set the oscilloscope to indicate the approximative 8-10 TP-4 clock pulses.
3. Plot waveform of TP-1 based on clock waveform. And then try to discriminate each of Ch1-Ch4 from the waveform PAM multiplexed shown in the DC input level of each channel and TP-1.



4. Connect Y-1 probe of Oscilloscope to TP-2 and Y-2 probe to TXOUTPUT terminal. Plot corresponding waveform and find Bit row corresponding to the value quantized according to the DC input of each channel, from the waveform in this Chart.



5. Set clock switch to SLOW this time. While performing the above procedure 4 again.

6. Through the above experiments, the process where sampled and quantized levels by time division are converted into digital values by AD converter may be understood. Represent the relationship of SOC (start of Conversion) of TP-2 and Tx Output in a time Chart.

7. Connect U-2990F and U-2990G Unit and connect Test Tones in transmitting part to each input amplifier. Then set clock switches in transmitting/receiving parts to FAST.

8. Connect Y-1 and Y-2 probe of oscilloscope to Amp. input and output of CH-1 output respectively, and turn the power switch ON. Perform setting of oscilloscope so that 350 Hz sinusoidal waveform of CH-1 output may be displayed on the screen.

9. Plot the waveform displayed on the oscilloscope screen through the above procedure, based on Y-1 probe waveform which is in the status before passing LPF and output waveform of Y-2 probe is demodulation waveform which has been restored by passing Low Pass Filter, Confirm whether same frequencies as in transmitting input (700 Hz, 1400 Hz and 2800 Hz) is being received.

DISCUSSION

- 1- Which multiplexing is used in PCM?
- 2- How is frame synchronization achieved in PCM TDM system?
- 3- Where is TDM used?
- 4- What are the advantages of TDM?